H-TABLE

A PROJECT REPORT

SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENT FOR THE AWARD OF THE DEGREE OF

B.TECH. (COMPUTER ENGINEERING)
TO
RK UNIVERSITY, RAJKOT

SUBMITTED BY

Name of Student	Enrollment No.
Duragiya Hitesh Vasudev	20SOECE13007

UNDER THE GUIDANCE OF

Internal Guide	**External Guide**
Chhaya Patel	Ronak Gandhi
Assistant Professor,	Project Coordinator
RK University,	RKIT Software Private Limited
Rajkot	Rajkot-Gujarat

April 2023

SCHOOL OF ENGINEERING, RK UNIVERSITY, RAJKOT

DECLARATION

I hereby certify that I am the sole author of this project work and that neither any part of this project work nor the whole of the project work has been submitted for a degree to any other University or Institution. I certify that, to the best of my knowledge, my project work does not infringe upon anyone's copyright nor violate any proprietary rights and that any ideas, techniques, quotations, or any other material from the work of other people included in my project document, published or otherwise, are fully acknowledged in accordance with the standard referencing practices. I declare that this is a true copy of my project work, including any final revisions, as approved by my project review committee.

Signature of Student

Duragiya Hitesh Vasudev (20SOECE13007),
Date: 29/04/2023
Place: Rajkot

INDUSTRY CERTIFICATE

RKIT
SOFTWARE PVT. LTD.

Date: 13-04-2023

<u>Subject – Internship Completion Letter</u>

Dear Mr. Hitesh Duragiya,

We are pleased to certify you for the successfully ongoing College Internship as a Full Stack Developer at our company.

During the internship period from 1st August,2022 to 30th June,2023, you have been maintaining the positive approach toward the assigned tasks.

We appreciate the sincerity and motivation shown. We are grateful that you have started your professional career with us.

RKIT Software Pvt RAJKOT

For,

RKIT Software Pvt Ltd

Miracle® Accounting Software
RKIT Software Pvt Ltd
R K House, 4/11 - Bhaktinagar Station Plot, Rajkot 360 002. INDIA
: (0281) 6161400 : sales@rkitsoftware.com : www.rkitsoftware.com

CERTIFICATE

This is to certify that the work which is being presented in the Project Report entitled **"H-TABLE",** in partial fulfilment of the requirement for the award of the degree of **B.Tech. (Computer Engineering)** and submitted to the School of Engineering, RK University, is an authentic record of my own work carried out during a period from **August 2022 to April 2023.** The matter presented in this Project Report has not been submitted by me for the award of any other degree elsewhere.

Signature of Student

Duragiya Hitesh Vasudev (20SOECE13007),

This is to certify that the above statement made by the student is correct to the best of my knowledge.

Internal Guide	**External Guide**	**Head of Department**
Chhaya Patel	Ronak Gandhi	Dr. Nirav Bhatt
Assistant Professor,	Project Coordinator,	CE / IT / BCA / MCA
RK University,	RKIT Software Private Limited,	School of Engineering,
Rajkot	Rajkot-Gujarat	RK University, Rajkot

April 2023

SCHOOL OF ENGINEERING, RK UNIVERSITY, RAJKOT

ACKNOWLEDGEMENT

It has indeed been a great privilege for me to have **Mr. Ronak Gandhi**, Project Coordinator of RKIT Software, Rajkot, and **Mrs. Chhaya Patel**, Assistant Professor of RK University, as my mentors for this project. His awe-inspiring personality, superb guidance, and constant encouragement are the driving forces behind this project. I take this opportunity to express our utmost gratitude to him. I am also indebted to him for his timely and valuable advice.

I am highly grateful to all the technical and non-teaching staff of the Development Department of RKIT Software and Computer Engineering Department of RK University for their constant assistance and cooperation.

ABSTRACT

This project aims to develop an automatic dynamic grid system that can adjust to the size and content of the website or application, providing an optimal layout for the user interface. The system will calculate the best grid configuration based on the URL entered in the textbox, and a grid will be displayed. The grid system will also automatically adapt to changes in the content, reorganising the layout without requiring manual intervention.

The proposed system will be implemented using HTML, CSS, and JavaScript and will include an intuitive user interface to allow for easy customization of the grid settings. The system will be designed to be flexible and scalable, allowing it to be integrated into various web applications and content management systems. The project's expected outcomes include the development of a functional automatic dynamic grid system that will provide an efficient and optimised layout for websites and applications. This system will eliminate the need for manual adjustments and reduce the time and effort required for layout design. Additionally, the system will improve the overall user experience by providing a more organised and visually appealing interface.

Overall, this project aims to improve grid layout design and provide an effective solution for creating dynamic and adaptive interfaces.

TABLE OF CONTENT

TABLE OF CONTENT

COMPANY PROFILE

Name of Organization:	RKIT Software Private Limited
Managing Director:	Dhimant Patel
HR/Contact Person Name:	Sejal Kothari
Phone No.: 0281 616 1400	Mobile No.: +91 99749 96496
Email Address:	hrd@rkitsoftware.com
Industrial supervisor's name:	Ronak Gandhi
Designation:	Project Coordinator
Phone No.: -	Mobile No.: +91 84018 28562
Email Address:	ronak.gandhi@rkitsoftware.com
Year of Establishment:	1998
Address:	RK House, 4/11, Gondal Road, Bhakti Nagar Station Plot, Rajkot, Gujarat 360002.
Product Details:	Miracle Accounting Software
Details of testing facilities available:	-
Any other special information:	-

COMPANY INFRASTRUCTURE

COMPANY INTRODUCTION

Infrastructure at RKIT Software Pvt Ltd - friendly environment, our team provides value-focused advice on client's site. We are helping clients to determine what to build, our team knows how to do it more quickly and cost efficiently. RKIT Software Pvt. Ltd. was started in 1998 by three young entrepreneurs. Today we are amongst the top 10 accounting software providers in India.

Initially, the company started the development of customized software and gradually shifted to different kinds of Packaged Accounting Software. General-purpose accounting software developed by the company is very popular in the market and brand leader in their respective areas with more than 75,000 users.

Company's motto is (One team one Vision) in the development of software packages is 'Keep It Easy & Advance'. Users feel comfortable while using our products and do not require any special skills to use it.

VISION

RKIT Software Pvt Ltd India based Accounting Software company and we are offering reasonably priced software (Miracle), development and promotional services to allow presence

1.0 INTRODUCTION

1.1 PROJECT SUMMARY

A dynamic grid system automatically adjusts the layout of a website or application based on the data. The system calculates the optimal grid configuration using URL data and can adapt to changes in the content without requiring manual intervention. It is typically implemented using HTML, CSS, and JavaScript and can be integrated into various web applications and content management systems.

The benefits of a dynamic grid system include improved efficiency and quality of web design, as well as a more organised and visually appealing interface for the end user. It eliminates the need for manual adjustments, reducing the time and effort required for layout design. Dynamic grid systems have practical applications in a range of contexts, including e-commerce, social media, and content-heavy websites.

1.2 PURPOSE

- The user enters the URL into a textbox, and the grid system generates it automatically.
- The user can manipulate the data based on the requirement, like insert, update, or delete data.
- The user can change the data with stylish grid systems.
- The user can change the data according to their needs with filtering options.
- Also, users can export the data to an Excel sheet.

1.3 SCOPE

The scope of the project is to allow users to enter URLs that provide data only in simple arrays or simple JSON-format data.

1.4 TECHNOLOGY AND LITERATURE REVIEW

1.4.1 Technology Review:

Dynamic grid systems are typically implemented using HTML, CSS, and JavaScript. The HTML provides the structure for the grid, while the CSS defines the grid's appearance and behaviour. JavaScript is used to handle dynamic content and enable the grid to adjust its layout automatically.

1.4.2 Literature Review:

Dynamic grid systems have been widely studied and discussed in the field of web design. Research has shown that dynamic grids can improve the efficiency and quality of web design by automating the layout design process and reducing the need for manual adjustments

2.0 PROJECT MANAGEMENT

2.1 PROJECT PLANNING AND SCHEDULING

2.1.1 Project Development Approach

- Pre-Project Preparation: During this phase, the team defines the project scope, objectives, and requirements and creates a high-level project plan.
- Sprint Review: At the end of each module iteration, I gather feedback from my colleagues and seniors.
- Continuous Improvement: Throughout the project, I continuously refine and improve the process to ensure the project is built on time and to the required quality standards.

2.1.2 Project Plan

The project plan goes like this:

- Competitor Analysis
- Laying Out a Workflow Process
- Tracking Time
- Continuous Iterating and Testing

2.2 RISK MANAGEMENT

2.2.1 Risk Identification

- Data Privacy and Security Risks
- Human Error
- Incompatible URLs

2.2.2 Risk Analysis

- **Security vulnerabilities:** JavaScript-based dynamic grid systems may be vulnerable to various types of cyber threats, such as cross-site scripting (XSS) attacks, SQL injection attacks, and phishing attacks. These vulnerabilities could allow attackers to compromise sensitive data or take control of the system.

- **Performance issues:** Dynamic grid systems that rely heavily on JavaScript may experience performance issues, such as slow page load times, high CPU usage, or memory leaks. These issues could result in reduced usability or even system crashes.

- **Compatibility issues:** Dynamic grid systems that use JavaScript may have compatibility issues with different browsers or operating systems. This could result in inconsistencies in the user interface or functionality.

- **Data quality issues:** Dynamic grid systems that rely on JavaScript for data input or validation may be susceptible to data quality issues, such as invalid or incomplete data, duplicate data, or inconsistent data.

2.2.3 Risk Planning

- **Identify potential risks:** Begin by identifying all the potential risks that your dynamic grid system could face. These could include technical risks such as compatibility issues, security risks, and performance risks such as slow load times.

- **Assess risks:** Once you have identified the potential risks, assess each risk to determine its likelihood and potential impact. This will help you prioritise which risks addressing first.

- **Mitigate risks:** Develop a plan for mitigating each risk. This might involve taking steps such as implementing security measures, optimising performance, or creating backup systems in case of failure.

- **Monitor risks:** Even after you have taken steps to mitigate risks, you should continue to monitor your dynamic grid system for potential issues. This will help you catch problems early and address them before they become major issues.

- **Continuously improve:** Finally, continue to improve your risk management plan over time. Regularly evaluate your system for new risks and update your plan accordingly.

3.0 SYSTEM REQUIREMENT STUDY

3.1 HARDWARE AND SOFTWARE REQUIREMENT

Minimum Hardware Requirement

- Hard Disk: 500 MB of Free Space
- RAM: 1 GB.
- CPU: Intel Dual Core
- Windows 7 or later (32-Bit,64-bit), x86-64 based.

Minimum Software Requirement

- Visual Studio Code
- Web browser (Any)

4.0 SYSTEM ANALYSIS

4.1 STUDY OF CURRENT SYSTEM

The current system of system analysis for dynamic grid systems involves using various techniques and tools to understand and optimize the performance of these systems. Here are some of the key approaches and tools used in the analysis of dynamic grid systems:

- **Performance profiling:** This involves using tools like Chrome DevTools to measure the performance of dynamic grid systems in terms of rendering time, memory usage, and other metrics.

- **Code review:** This involves examining the code used to create dynamic grid systems to identify potential performance issues, code smells, and other areas for optimization.

- **Automated testing:** This involves creating automated tests that simulate user interactions with dynamic grid systems to identify potential bugs, errors, and other issues.

- **Optimization techniques:** This includes using techniques like lazy loading, caching, and code splitting to optimize the performance of dynamic grid systems.

4.2 PROBLEM AND WEAKNESS OF THE CURRENT SYSTEM

This system is not fully stable and does not have community support. There might be some bugs and loopholes in the system.

4.3 REQUIREMENT OF NEW SYSTEM

Add more functionality to the existing system, such as allowing users to enter URLs, and the grid system will generate automatically based on the URLs data.

4.4 FEASIBILITY STUDY

A feasibility study of a dynamic grid system involves assessing whether it is technically and financially feasible to implement such a system.

- **Technical feasibility:** This involves assessing whether the necessary technologies and skills are available to implement a dynamic grid system. This includes evaluating the availability of web development frameworks and libraries as well as the skills of the development team.

- **Market demand:** This involves assessing whether there is sufficient demand for a dynamic grid system among potential users, such as web developers and designers.

- **Cost analysis:** This involves estimating the costs of developing and maintaining a dynamic grid system, including the costs of hardware, software, development time, and ongoing maintenance.

- **Scalability:** This involves assessing whether the dynamic grid system can scale to accommodate larger datasets and more complex user interfaces.

- **Compatibility:** This involves assessing whether the dynamic grid system is compatible with different web browsers, operating systems, and devices.

- **Security:** This involves assessing the security of the dynamic grid system, including protection against hacking, data breaches, and other security threats.

4.5 FUNCTION OF THE SYSTEM

- Easy-to-use.
- Add, delete, and update data according to the user's needs.
- The user can use different and custom filters according to their requirements.
- Easily Sync Between Desktop Application.

5.0 TESTING

5.1 TESTING PLAN

When testing a dynamic grid system, it's important to have a comprehensive plan in place to ensure that all aspects of the system are thoroughly tested. Here is a possible testing plan:

- **Unit testing:** Begin with unit tests for individual functions and modules that make up the grid system. This includes testing for expected input and output, edge cases, and error handling.

- **Integration testing:** Once individual components have been tested, move on to integration testing. Test the interactions between different components and modules to ensure they work together as intended.

- **User interface testing:** Test the user interface to ensure that the grid system behaves correctly on different screen sizes and with different input methods (e.g., mouse vs touch). Test for usability, accessibility, and responsiveness.

- **Performance testing:** Test the performance of the grid system by measuring how quickly it loads, how it handles large amounts of data, and how it performs on older devices or slower networks.

- **Compatibility testing:** Test the grid system on different browsers and operating systems to ensure that it works as expected and that there are no compatibility issues.

- **Security testing:** Test the grid system for security vulnerabilities, such as SQL injection or cross-site scripting attacks.

5.2 TESTING STRATEGY

When it comes to testing a dynamic grid system, there are several testing strategies that can be employed to ensure its reliability and effectiveness. Here are some of the most common testing strategies for dynamic grid systems:

- **Functional Testing:** This strategy involves testing the different functionalities of the dynamic grid system to ensure that they are working as expected. It includes testing basic functionality, such as adding or removing rows and columns, as well as advanced functionality, such as filtering, sorting, and pagination.

- **Cross-Browser Testing:** Since different browsers can display web pages differently, it is important to test the dynamic grid system on different browsers to ensure that it works correctly on all platforms. This involves testing on popular browsers such as Chrome, Firefox, Safari, and Internet Explorer.

- **Load Testing:** This strategy involves testing the dynamic grid system to determine how well it performs under different load conditions. It includes testing for peak user loads, database performance, and server performance.

- **Regression Testing:** Whenever changes are made to the dynamic grid system, regression testing should be performed to ensure that the new features and updates have not introduced any bugs or issues. It involves re-testing the previously tested features to ensure that they still work as expected.

- **Security Testing:** This strategy involves testing the dynamic grid system to ensure that it is secure from common web application attacks such as SQL injection and cross-site scripting.

5.3 TESTING CASES

When testing a dynamic grid system, there are several test cases that should be considered to ensure that the system works as expected. Here are some common test cases for dynamic grid systems:

- **Adding and Removing Rows and Columns:** Test the ability to add and remove rows and columns dynamically. This includes testing for adding a single row or column, adding multiple rows or columns at once, and removing rows or columns.

- **Sorting:** Test the ability to sort data in ascending and descending order. This includes testing for single column sorting and multi-column sorting.

- **Filtering:** Test the ability to filter data based on specific criteria. This includes testing for filtering data based on one or more columns, as well as filtering data based on text or numeric values.

- **Pagination:** Test the ability to display a large amount of data by splitting it into multiple pages. This includes testing for the ability to navigate between pages, as well as testing for the display of the correct number of items per page.

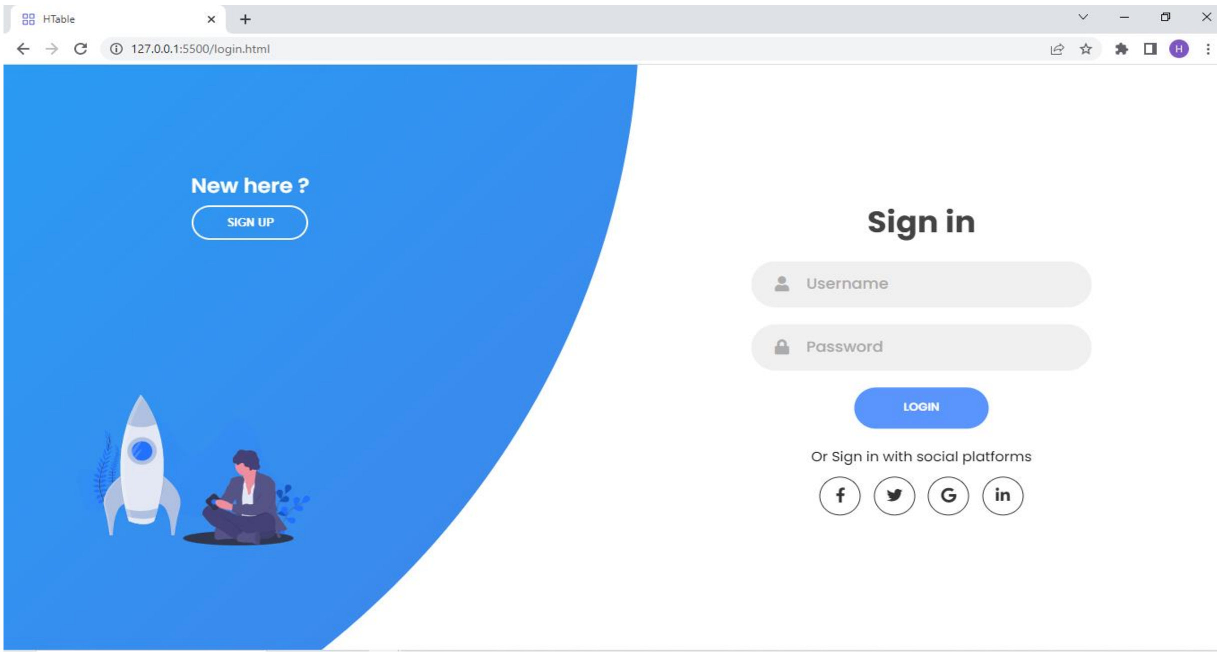
- **Selection:** Test the ability to select one or more rows or columns. This includes testing for selecting individual cells, rows, or columns, as well as testing for the ability to select multiple items at once.

- **Editing:** Test the ability to edit data in individual cells. This includes testing for editing text and numeric values, as well as testing for the ability to save or cancel changes.

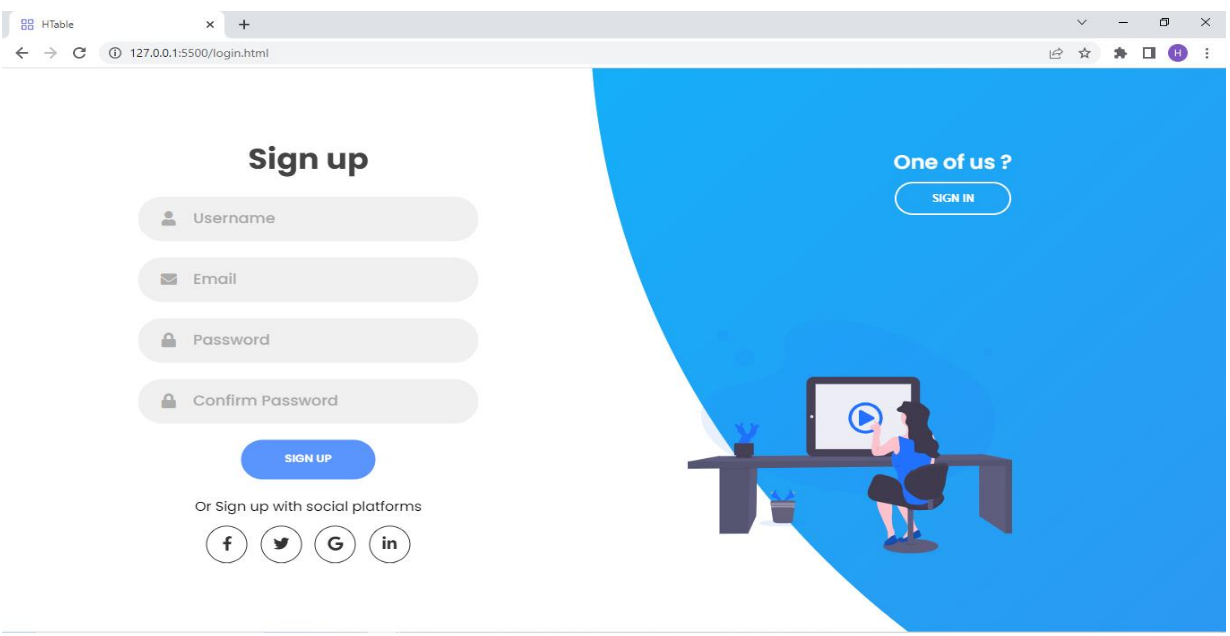
- **Data Validation:** Test the ability to validate data entered into the grid. This includes testing for validating text and numeric input, as well as testing for the display of error messages when invalid data is entered.

6.0 SCREEN SHOTS

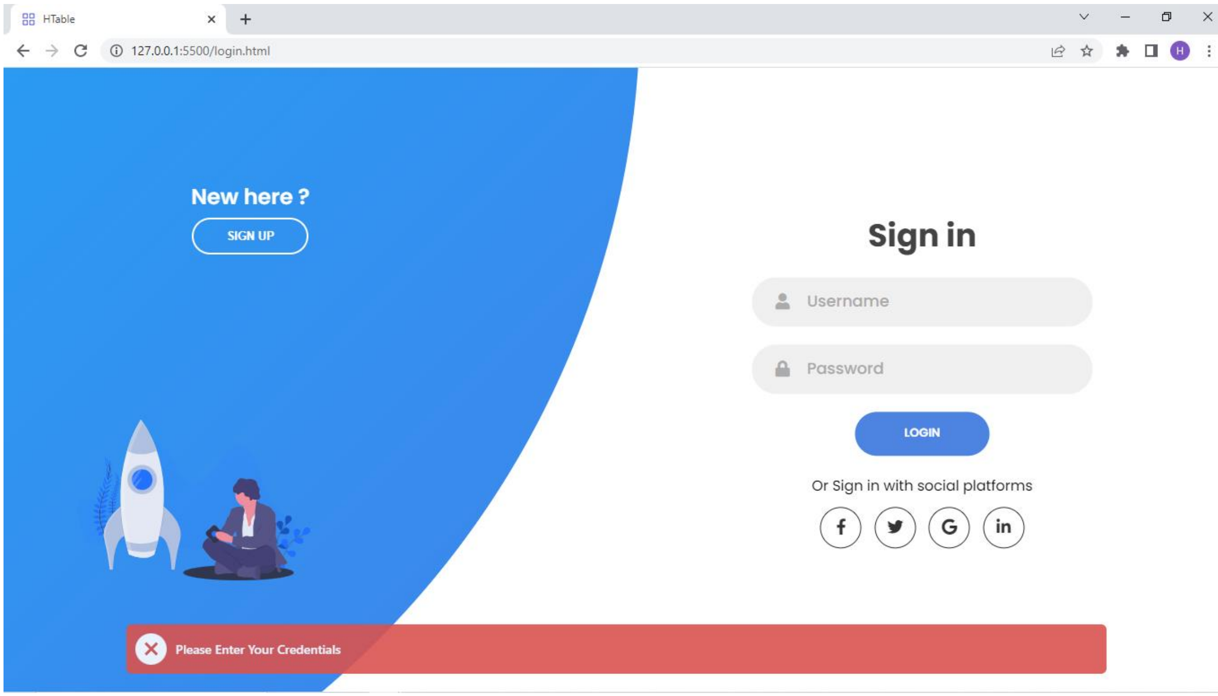
Sign in View



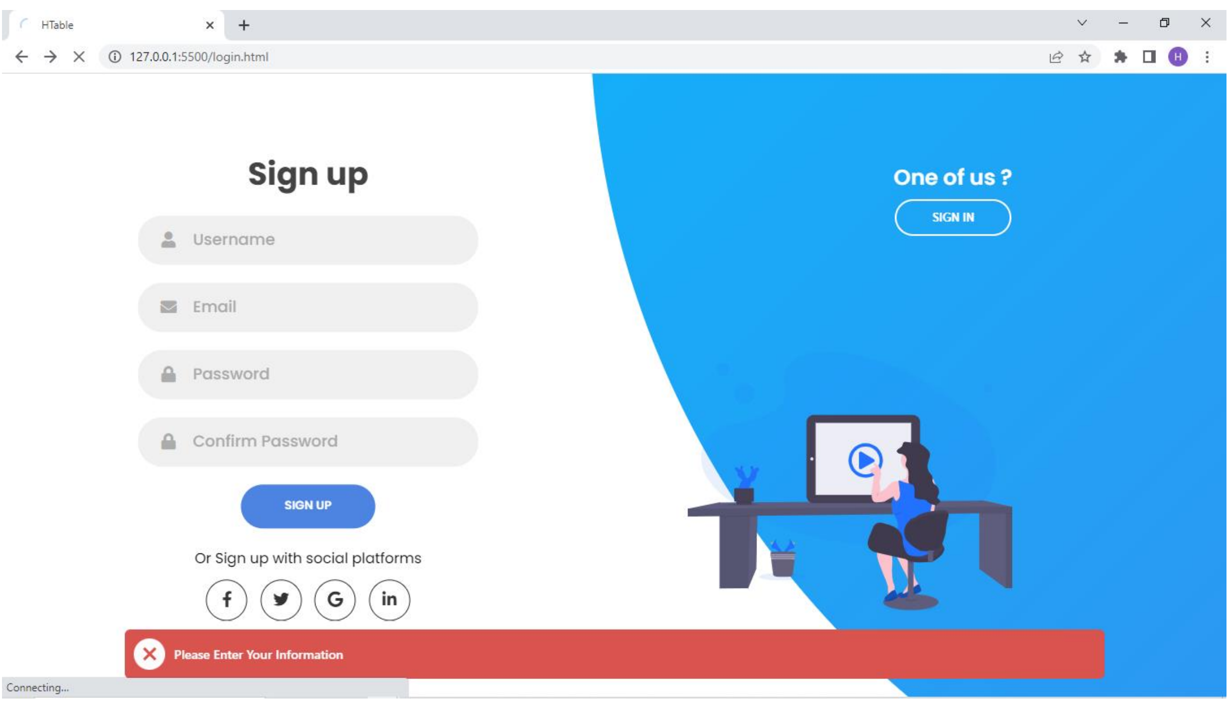
Sign up View



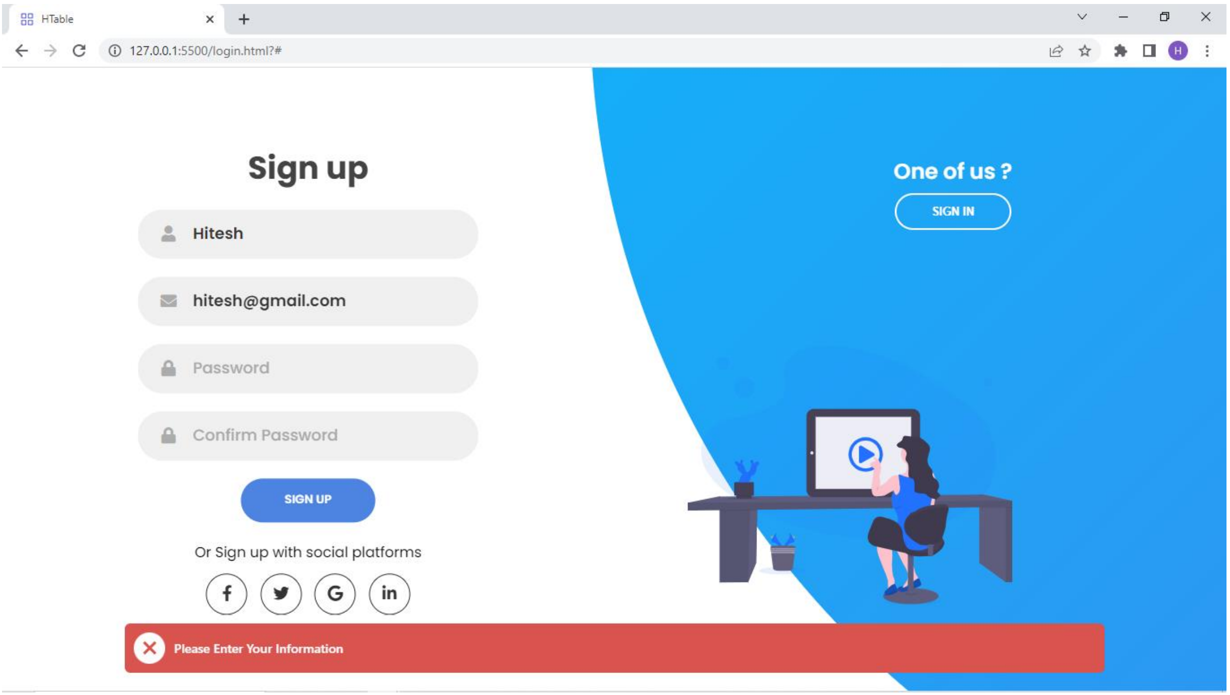
Validation on the sign-in page

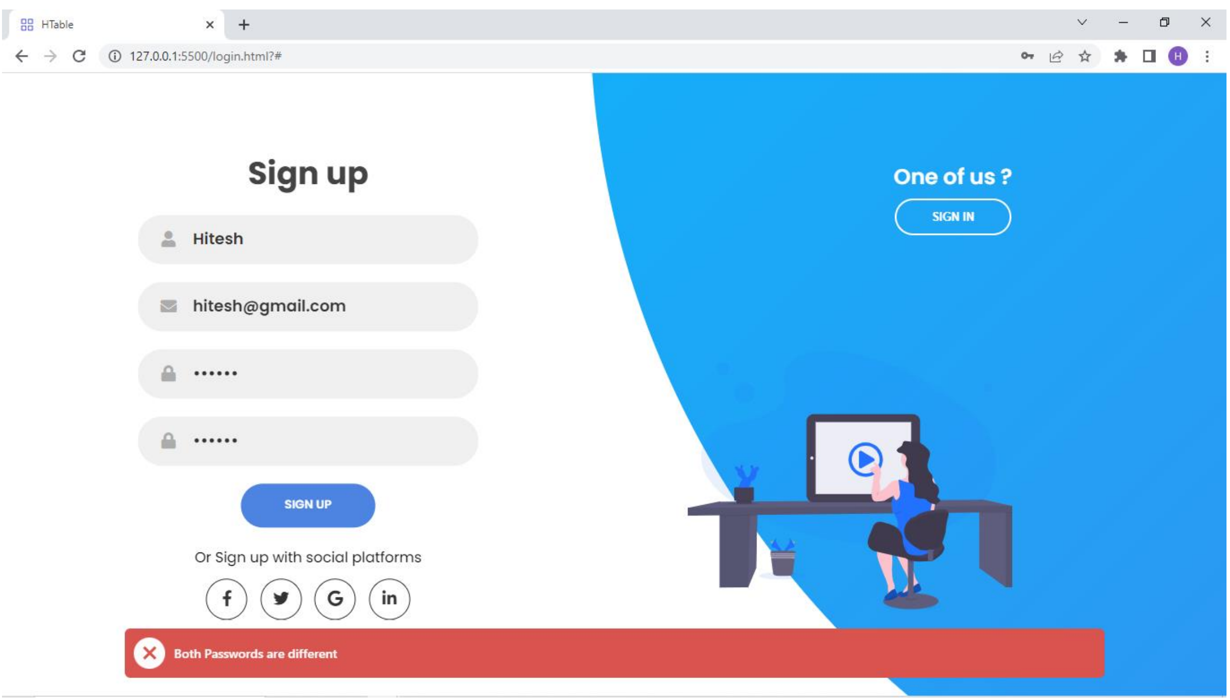


Validation on the sign-up page

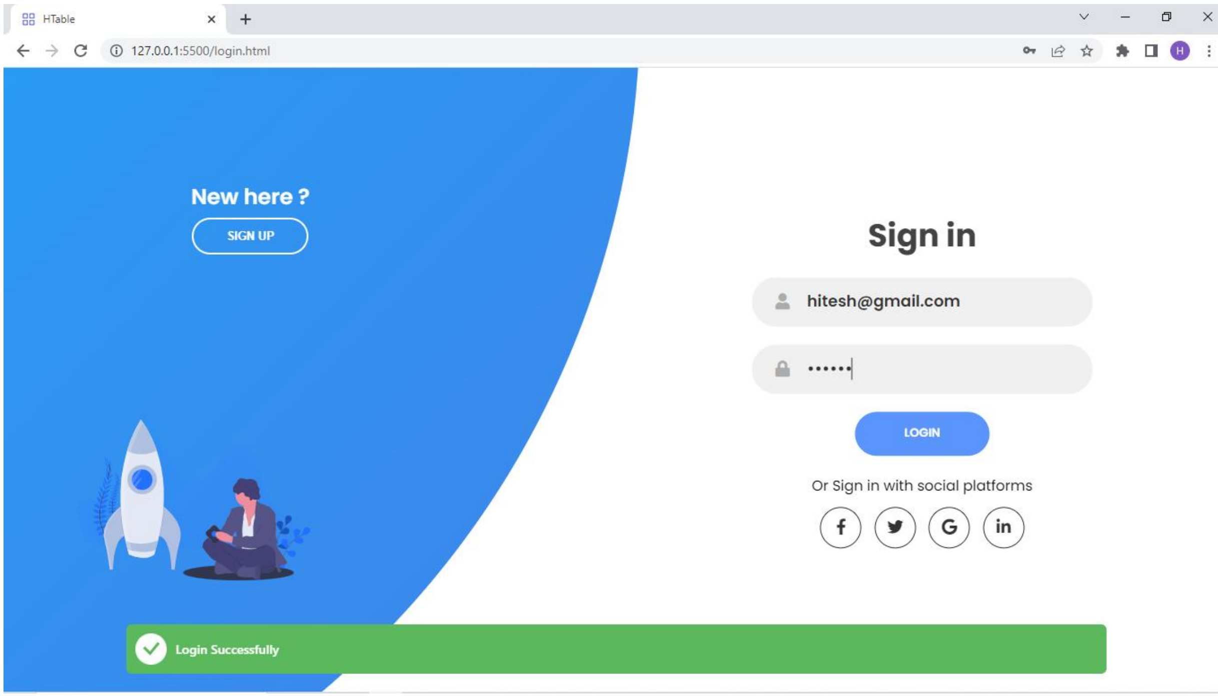


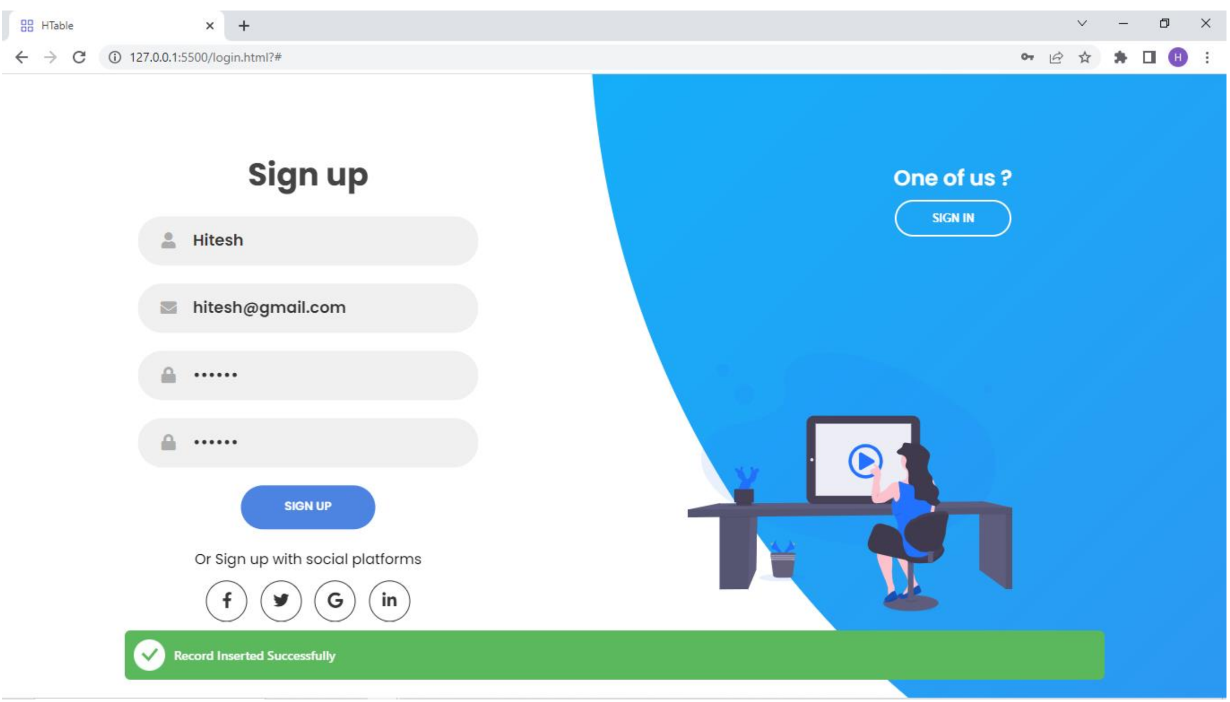
Validation on the sign-up page



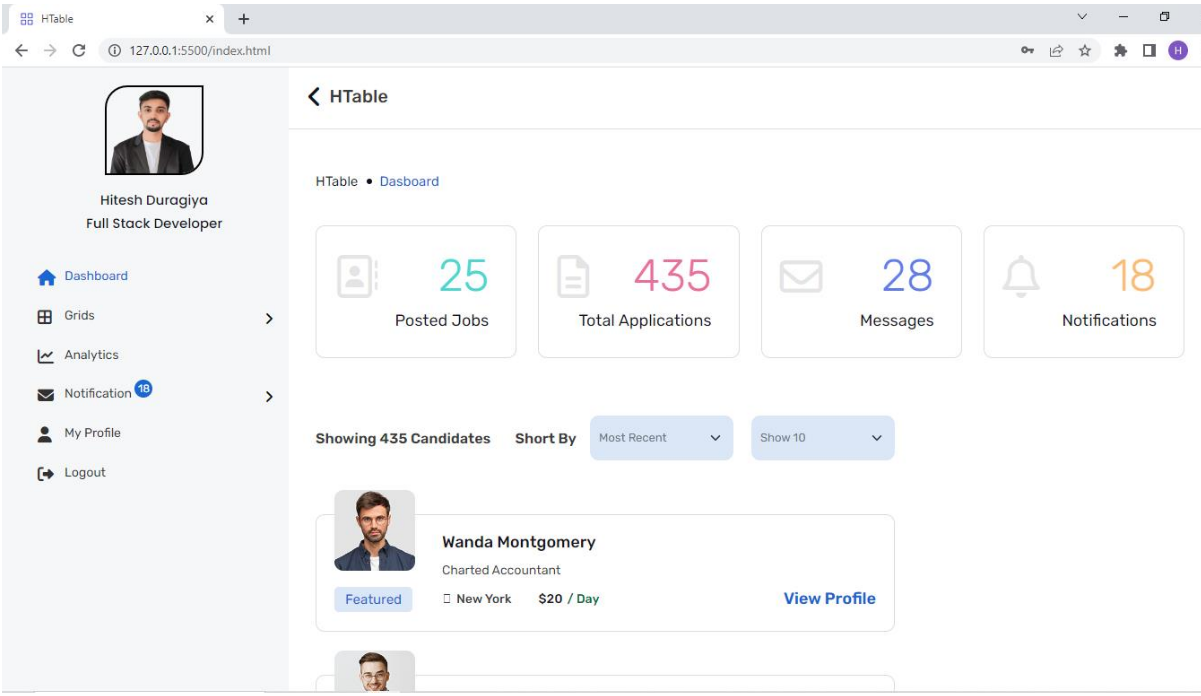


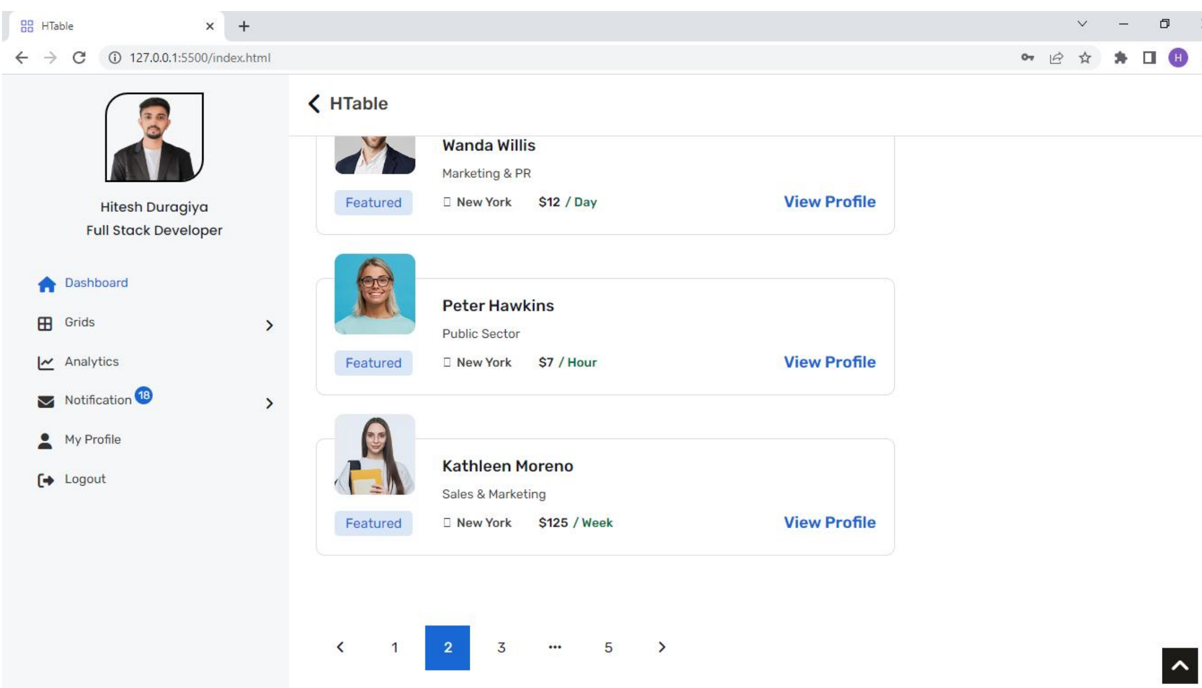
If User Entered Correct Credential or Information



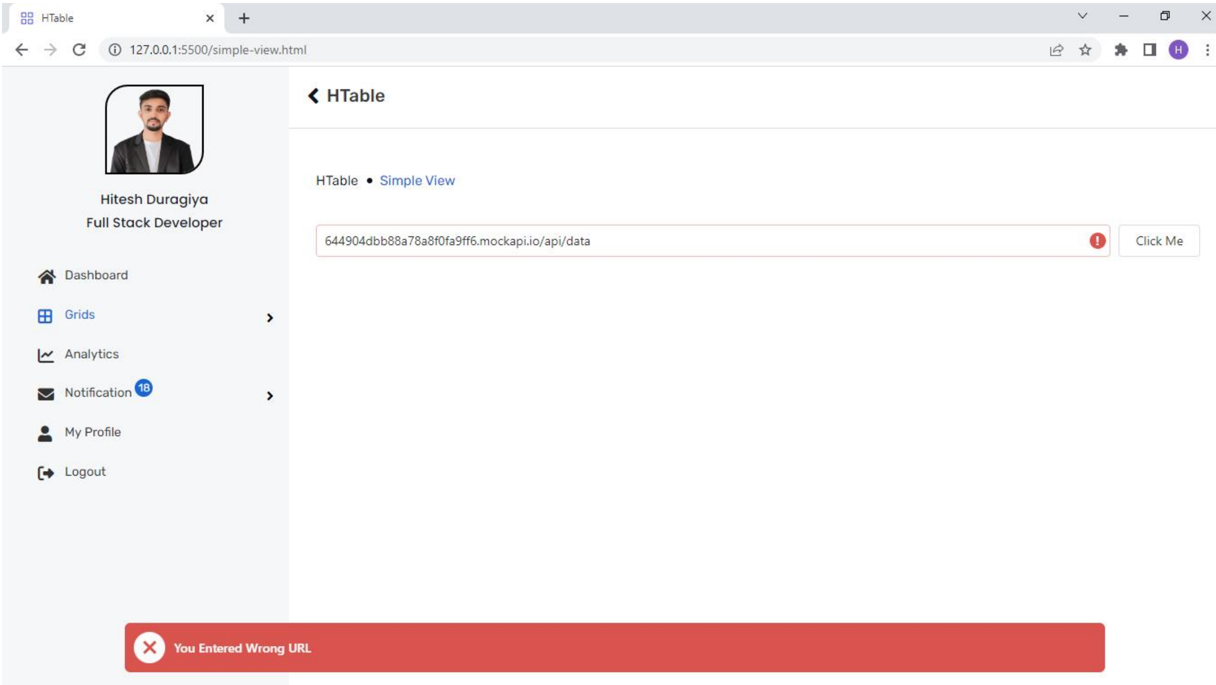


Dashboard View

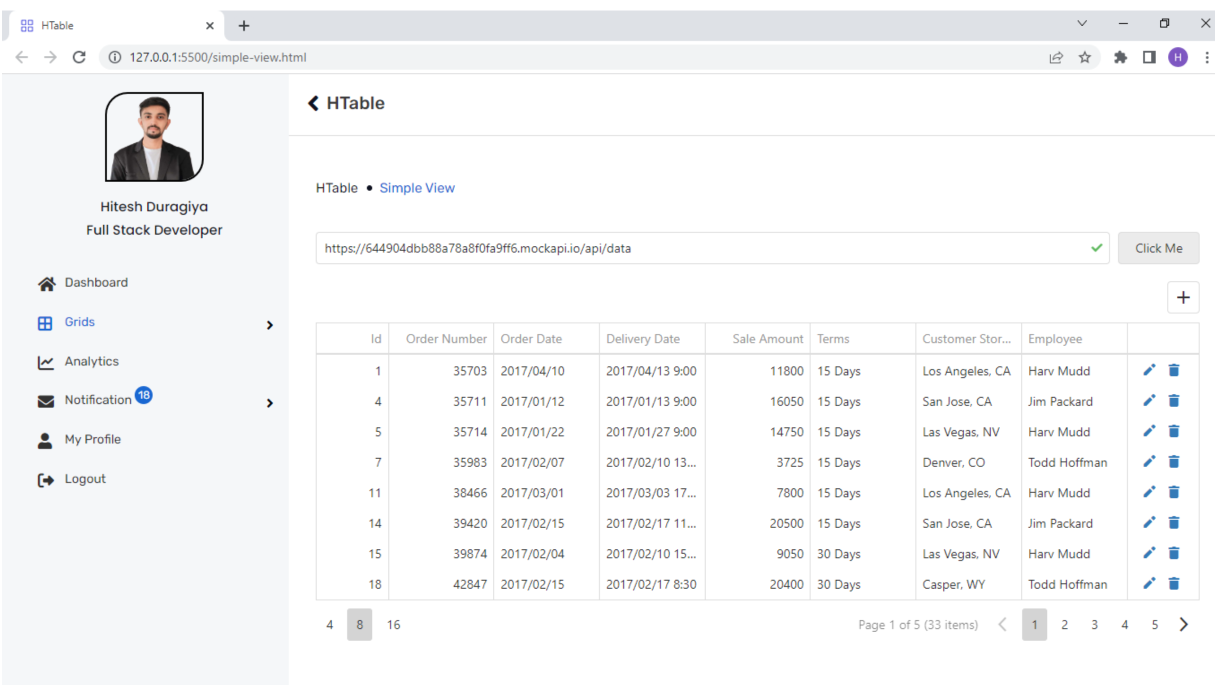




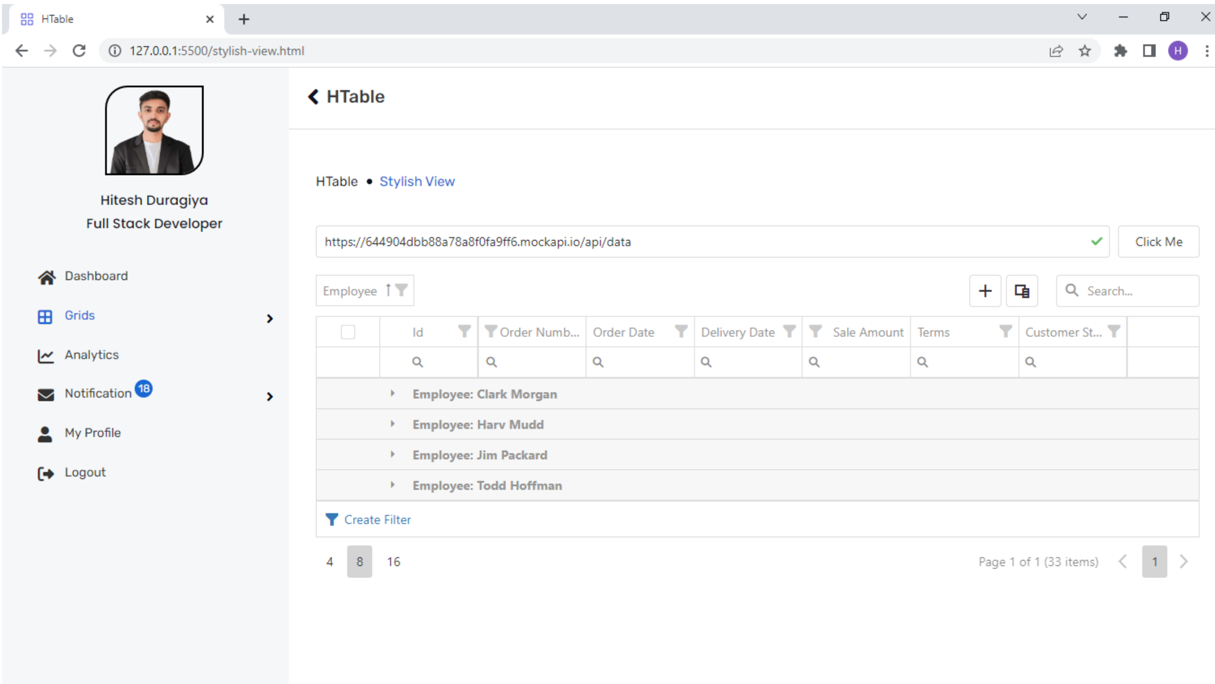
If User Entered Wrong URL



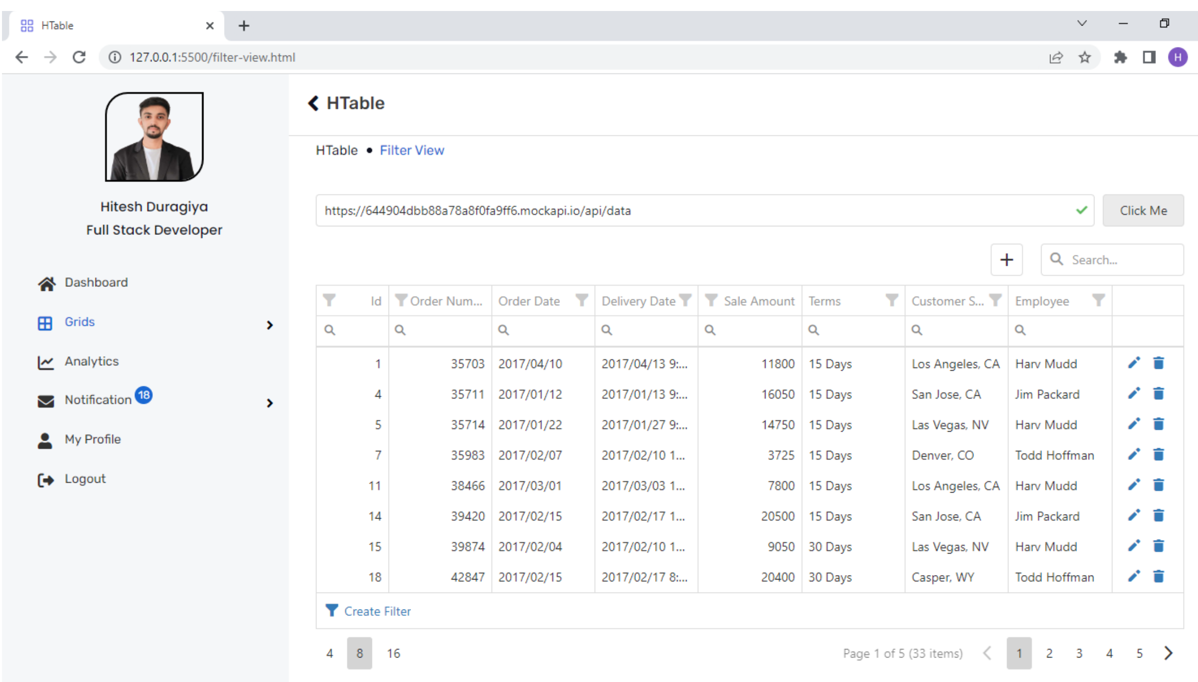
Dynamic Grid System with Simple View



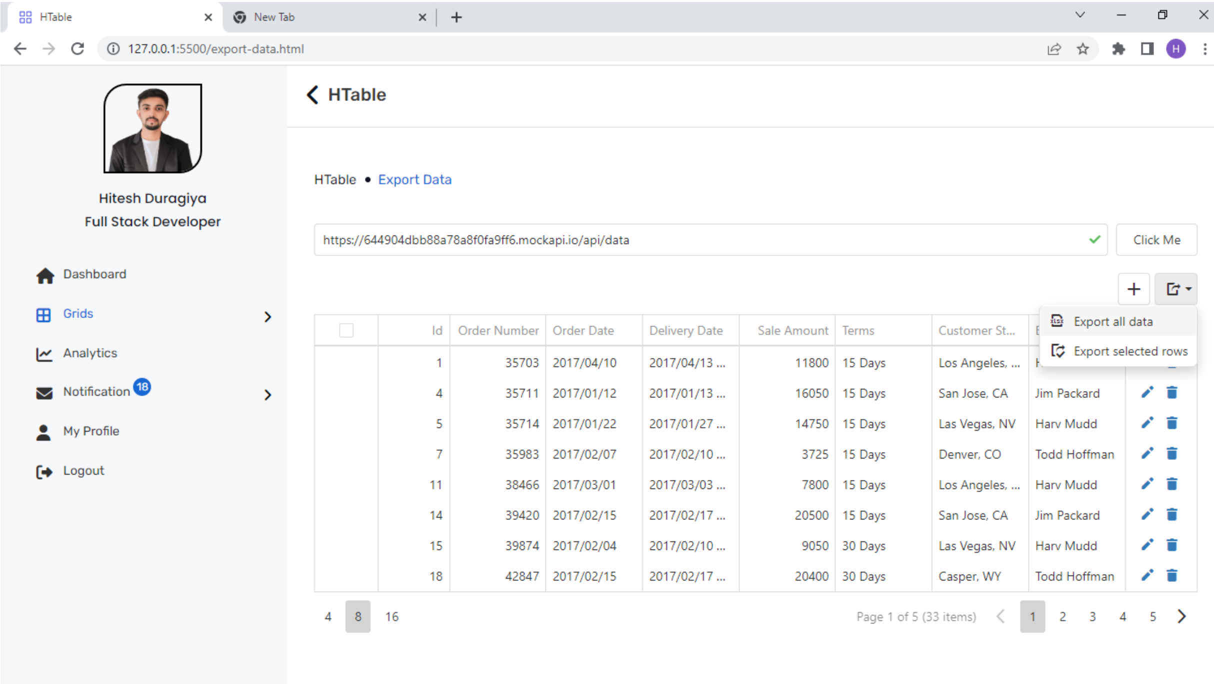
Dynamic Grid System with Stylish View



Dynamic Grid System with Filer View

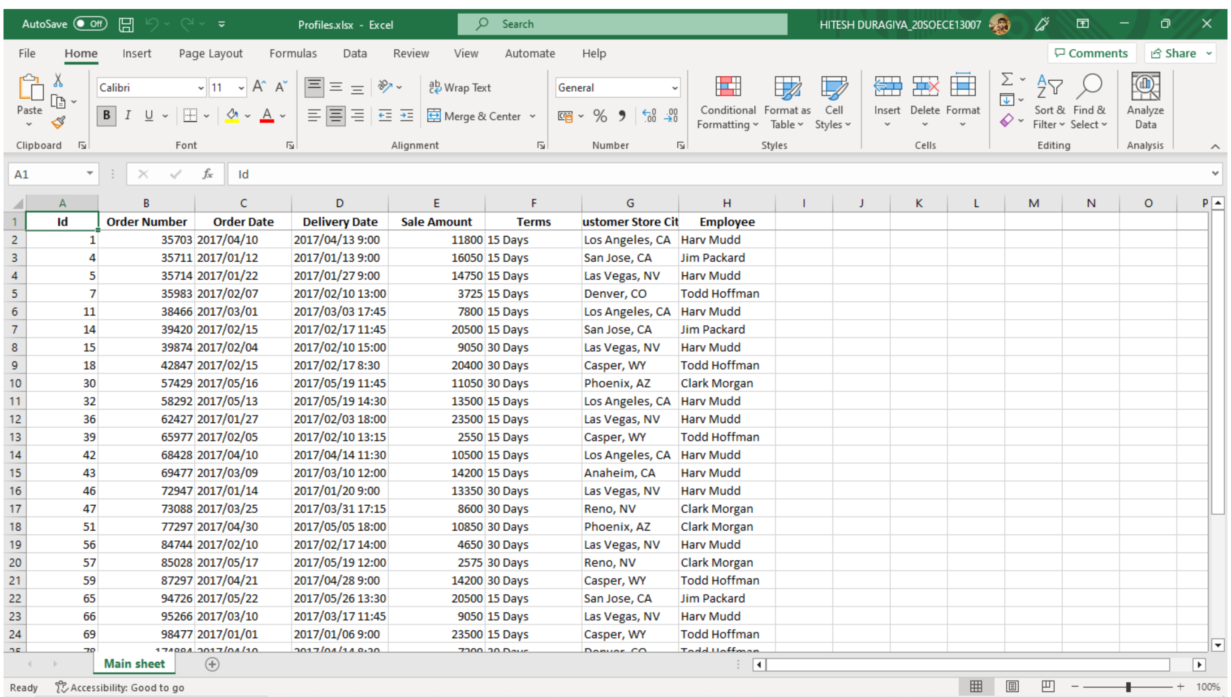


Dynamic Grid System with Export Facility



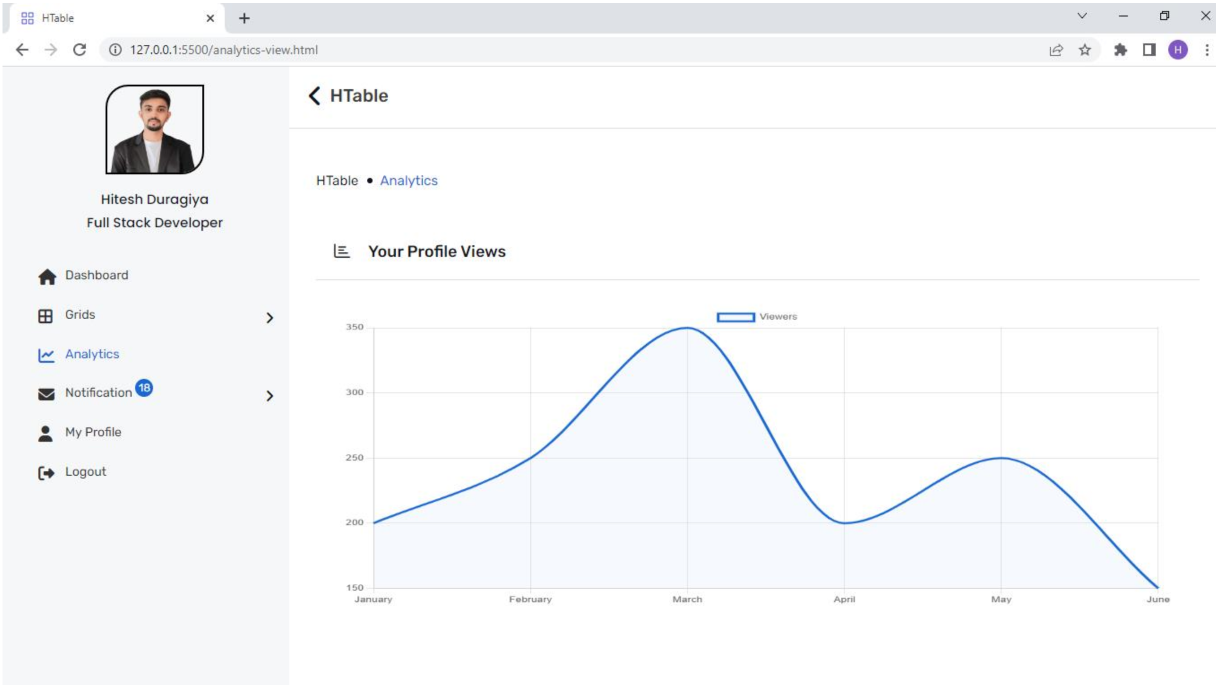
Id	Order Number	Order Date	Delivery Date	Sale Amount	Terms	Customer St...
1	35703	2017/04/10	2017/04/13 ...	11800	15 Days	Los Angeles, ...
4	35711	2017/01/12	2017/01/13 ...	16050	15 Days	San Jose, CA
5	35714	2017/01/22	2017/01/27 ...	14750	15 Days	Las Vegas, NV
7	35983	2017/02/07	2017/02/10 ...	3725	15 Days	Denver, CO
11	38466	2017/03/01	2017/03/03 ...	7800	15 Days	Los Angeles, ...
14	39420	2017/02/15	2017/02/17 ...	20500	15 Days	San Jose, CA
15	39874	2017/02/04	2017/02/10 ...	9050	30 Days	Las Vegas, NV
18	42847	2017/02/15	2017/02/17 ...	20400	30 Days	Casper, WY

Export Excel sheet

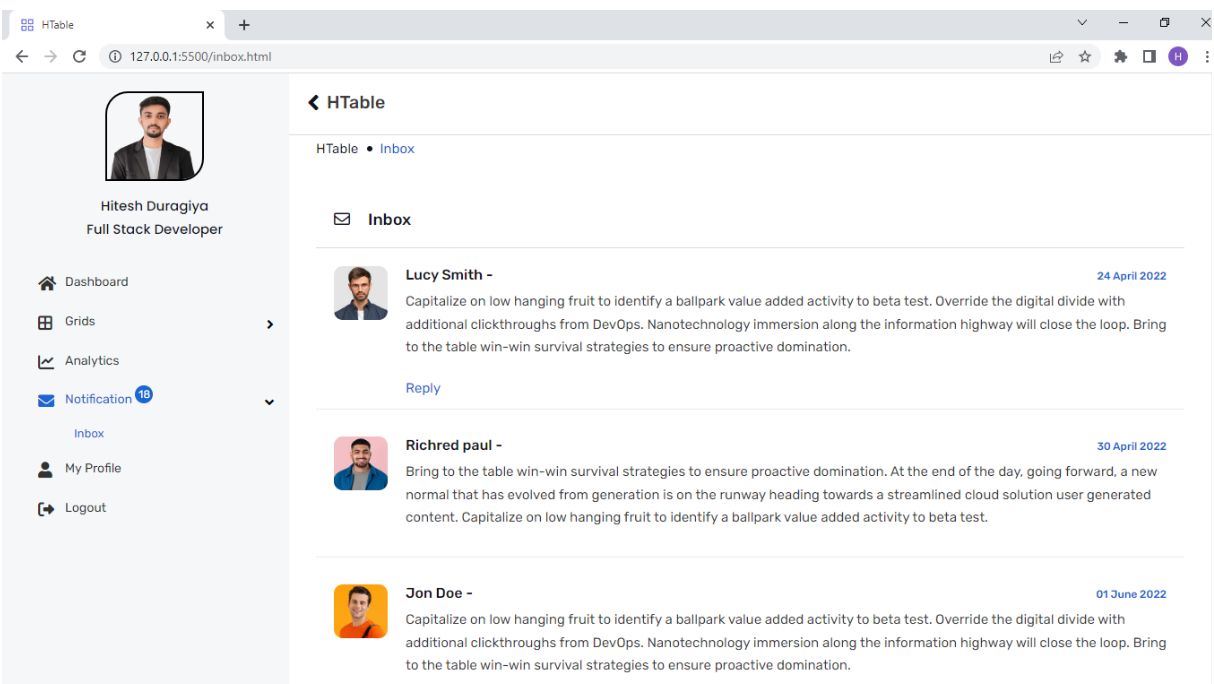


Id	Order Number	Order Date	Delivery Date	Sale Amount	Terms	ustomer Store Cit	Employee
1	35703	2017/04/10	2017/04/13 9:00	11800	15 Days	Los Angeles, CA	Harv Mudd
4	35711	2017/01/12	2017/01/13 9:00	16050	15 Days	San Jose, CA	Jim Packard
5	35714	2017/01/22	2017/01/27 9:00	14750	15 Days	Las Vegas, NV	Harv Mudd
7	35983	2017/02/07	2017/02/10 13:00	3725	15 Days	Denver, CO	Todd Hoffman
11	38466	2017/03/01	2017/03/03 17:45	7800	15 Days	Los Angeles, CA	Harv Mudd
14	39420	2017/02/15	2017/02/17 11:45	20500	15 Days	San Jose, CA	Jim Packard
15	39874	2017/02/04	2017/02/10 15:00	9050	30 Days	Las Vegas, NV	Harv Mudd
18	42847	2017/02/15	2017/02/17 8:30	20400	30 Days	Casper, WY	Todd Hoffman
30	57429	2017/05/16	2017/05/19 11:45	11050	30 Days	Phoenix, AZ	Clark Morgan
32	58292	2017/05/13	2017/05/19 14:30	13500	15 Days	Los Angeles, CA	Harv Mudd
36	62427	2017/01/27	2017/02/03 18:00	23500	15 Days	Las Vegas, NV	Harv Mudd
39	65977	2017/02/05	2017/02/10 13:15	2550	15 Days	Casper, WY	Todd Hoffman
42	68428	2017/04/10	2017/04/14 11:30	10500	15 Days	Los Angeles, CA	Harv Mudd
43	69477	2017/03/09	2017/03/10 12:00	14200	15 Days	Anaheim, CA	Harv Mudd
46	72947	2017/01/14	2017/01/20 9:00	13350	30 Days	Las Vegas, NV	Harv Mudd
47	73088	2017/03/25	2017/03/31 17:15	8600	30 Days	Reno, NV	Clark Morgan
51	77297	2017/04/30	2017/05/05 18:00	10850	30 Days	Phoenix, AZ	Clark Morgan
56	84744	2017/02/10	2017/02/17 14:00	4650	30 Days	Las Vegas, NV	Harv Mudd
57	85028	2017/05/17	2017/05/19 12:00	2575	30 Days	Reno, NV	Clark Morgan
59	87297	2017/04/21	2017/04/28 9:00	14200	30 Days	Casper, WY	Todd Hoffman
65	94726	2017/05/22	2017/05/26 13:30	20500	15 Days	San Jose, CA	Jim Packard
66	95266	2017/03/10	2017/03/17 11:45	9050	15 Days	Las Vegas, NV	Harv Mudd
69	98477	2017/01/01	2017/01/06 9:00	23500	15 Days	Casper, WY	Todd Hoffman

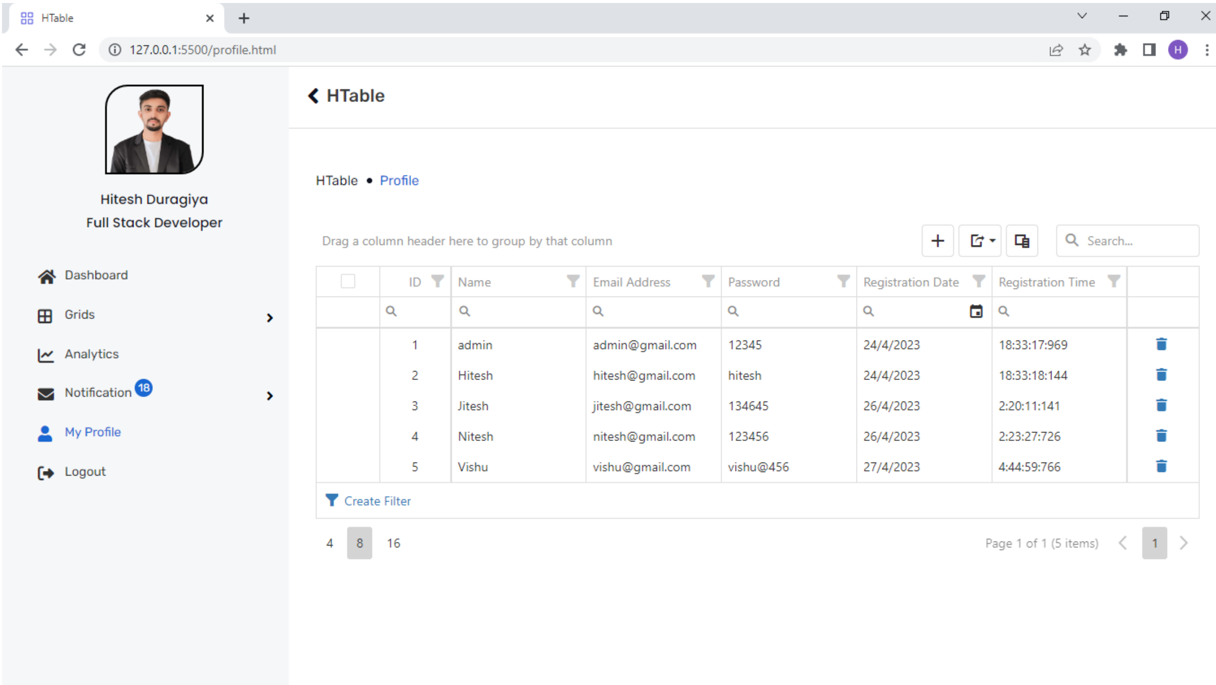
Analytics View



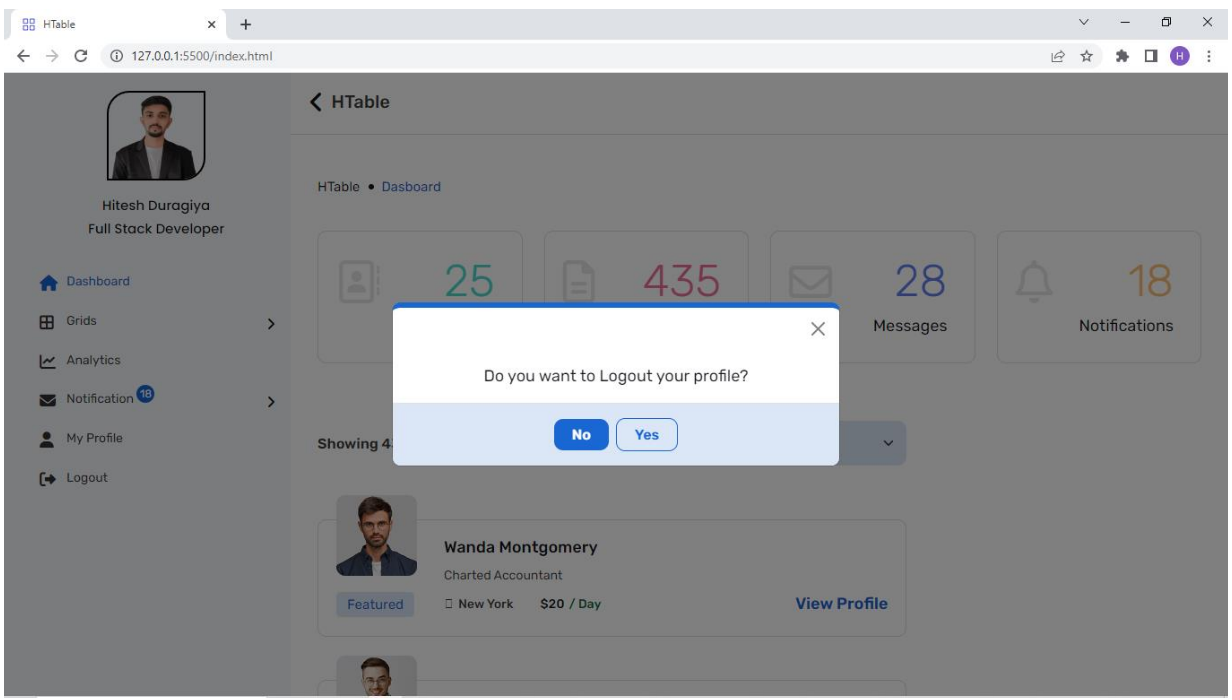
Notification View



Profile View



Logout View



7.0 LIMITATION AND FUTURE ENHANCEMENT

LIMITATIONS:

- **Performance Issues:** Large datasets can cause performance issues with dynamic grid systems, especially when sorting, filtering, and pagination features are used. The system may become slow and unresponsive, making it difficult for users to work efficiently.

- **Cross-browser Compatibility:** Dynamic grid systems may not work consistently across all web browsers, which can cause compatibility issues for users. This can result in an inconsistent user experience and reduced functionality.

- **Accessibility:** Dynamic grid systems may not be fully accessible to users with disabilities, which can limit their use for certain users. This can include issues with keyboard accessibility, screen reader compatibility, and colour contrast.

FUTURE ENHANCEMENTS:

- **Improved Performance:** Dynamic grid systems could be enhanced with features such as server-side pagination, caching, and lazy loading. These features can help improve the system's performance and make it more responsive to user input.

- **Enhanced Visualization:** Dynamic grid systems could be enhanced with more advanced visualization features, such as charts and graphs, to help users better understand and analyse their data. This would provide users with more powerful analysis capabilities and make the system more useful for data-driven applications.

- **Mobile Compatibility:** To enhance the accessibility of dynamic grid systems, they could be optimized for use on mobile devices. This would ensure that users can access and use the system on-the-go and provide a consistent user experience across devices.

- **Integration with other Technologies:** Dynamic grid systems could be enhanced by integrating with other technologies, such as data visualization tools, machine learning algorithms, and cloud-based data storage systems. This would provide more powerful functionality and make the system more useful for advanced applications.

REFERENCES

- **HTML:** **https://html.com**
- **CSS:** **https://developer.mozilla.org/en-US/docs/Web/CSS**
- **Bootstrap:** **https://getbootstrap.com**
- **JavaScript:** **https://www.javascript.com**
- **jQuery:** **https://jquery.com**
- **API:** **https://mockapi.io**